

YUEZHEN DONG

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RELEVANT SKILLS

Languages	Python, Java, TypeScript/JavaScript, C/C++, SQL, Ruby, Bash
ML/AI	PyTorch, scikit-learn, OpenAI API, Gemini API, RAG, LLM Agents, Prompt Engineering
ML Methods	Supervised learning, classification, model evaluation, causal inference, A/B testing
Frameworks	FastAPI, Flask, React, Node/Express, Ruby on Rails, Spring Boot, GraphQL
Infrastructure	Docker, PostgreSQL, AWS, Git, Linux

EDUCATION

University of Waterloo – Faculty of Engineering	Expected Graduation MAY 2028
BASc in Computer Engineering	Waterloo, ON, CAN

WORK EXPERIENCE

Software Engineering Intern	MAY 2026 – AUG 2026
Python, TypeScript, React Native, Ruby on Rails, RAG, LLM, OAuth, BLE	Shopify

- Contributed to **agentic AI tooling infrastructure** by implementing **RAG-based context retrieval** with embedding reranking and prompt orchestration pipelines, reducing irrelevant context fed to the LLM by **~40%** and improving AI-generated code acceptance rate in developer review.
- Built **Quick Device Activation** for Shopify POS using **BLE** pairing, QR/manual-code fallback, and **OAuth Device Authorization** – reducing merchant device setup time from **~5 minutes to under 30 seconds**.
- Instrumented login-conversion metrics across **React Native** and **Rails** using feature flags, enabling A/B testing of activation flows and identifying a drop-off step that accounted for **35% of incomplete activations**.

Software Engineering Intern	SEP 2025 – DEC 2025
React, TypeScript, Ruby on Rails, GraphQL	Shopify

- Owned a **staff assignment modal** in Retail Admin end-to-end – built React components and designed **GraphQL** queries with server-side pagination, reducing staff search query response time by **~60%**.
- Shipped full-stack features across **6 systems** (Admin Web, POS Mobile, Shopify Core, Business Platform, Cloud Sync Streamer, POS Channel), maintaining backward-compatible data contracts and achieving **zero regression** on launch.
- Collaborated with engineering, product, and design to clarify requirements and document tradeoffs, reducing mid-sprint scope changes and contributing to **on-time delivery of 3 consecutive** merchant-facing feature releases.

Web Developer Coop	JAN 2025 – APR 2025
Java, Spring Boot, Maven, Apache POI, Git	AGF Investments

- Built a **Java/Spring Boot** ScoreCard application with XLSX streaming via **Apache POI**, configurable transformation pipelines, and input validation, automating report generation across **4 departments** and saving **20+ staff hours/week** previously spent on manual spreadsheet processing.

PERSONAL PROJECTS

Multimodal Agent Evaluation Sandbox
Python, PyTorch, Gemini API, FastAPI, React, PostgreSQL, Docker

- Built a **multimodal benchmark for AI agents** evaluating text, image, code, and tabular reasoning through **tool-calling**, **RAG**, sandboxed Python execution, and verifier-based scoring – agents evaluated on this benchmark achieved **23% higher accuracy** on unseen task variants compared to a zero-shot baseline.

AI Merchant Inventory Assistant
Python, TypeScript, OpenAI API, scikit-learn, React, Node/Express, SQLite

- Built an **LLM-powered inventory assistant** with an **agentic tool-routing pipeline** that classifies merchant intent and dispatches to structured SQLite queries across 5 inventory workflows, achieving **89% intent classification accuracy** on a 200-query evaluation set.

3D Data Sandbox
Electron, Node.js, Three.js, PapaParse, JavaScript

- Built a cross-platform **Electron** desktop app using **Three.js** for offline 3D visualization of CSV/JSON datasets up to 100K rows; implemented **K-means** clustering, outlier detection, and interactive highlighting with **sub-200ms render updates** on dataset filter changes.